

Lei Shen

Wuhan, China | leishen@whu.edu.cn | <https://leishen00.github.io> | ORCID

EDUCATION

- **Wuhan University**, Wuhan, Hubei, China Sep 2023 – Present
M.S. in Medicine in Surgery, GPA: 3.71/4.0
Honors: National Scholarship for Graduate Students for two years (2023 & 2024)
- **Wuhan University**, Wuhan, Hubei, China Sep 2018 – May 2023
B.S. in Clinical Medicine, GPA: 3.58/4.0
Honors: Outstanding Graduate of Wuhan University (2023), Lei Jun Scholarship of Wuhan University (2022)

RELEVANT COURSEWORKS

- **Graduate-level:** Evidence-Based Clinical Medicine (A), Introduction to Clinical Research (A), Guided Reading of Medical Literature (A), Academic Ethics and Regulations (A), Surgery (B), Medical Professionalism (A), etc.
- **Undergraduate-level:** Basic Medicine Comprehensive Experiment (A), Clinical Pathophysiology and Therapeutics (A), Neuroscience (B), Surgical Internship (A), Internal Medicine Internship (A), etc.

SELECTED PROJECTS

- Project 1: The role of Cuproptosis in glioma progression** Mar 2022 - Jun 2023
Location: Wuhan University Lab; Advisor: Prof. Xiong
Cuproptosis is a novel type of programmed cell death. I designed and led an *National Training Program of Innovation and Entrepreneurship for Undergraduates* to investigate the role of cuproptosis in glioma progression.
- Applied bioinformatics pipelines to analyze glioma-related datasets for cuproptosis signatures.
 - Conducted *in vitro* cellular and molecular experiments to validate functional roles of cuproptosis.
- Project 2: Spatially multidimensional proteomics of glioma** Sep 2021 - Mar 2024
Location: Wuhan University Lab; Advisor: Prof. Xiong
As a core member, I contributed to a multi-cohort proteomic study that uncovered novel molecular mechanisms in diffuse glioma. This collaborative work was published in *Science Advances* as the first-author.
- Performed proteomic sample preparation and downstream data analysis.
 - Contributed substantially to data visualization and manuscript writing.
- Project 3: Functional mechanism of PEPD in glioblastoma** Mar 2024 - Present
Location: Wuhan University Lab; Advisor: Prof. Xiong
Building on previous proteomic discoveries, I focused on the role of PEPD in glioblastoma. I demonstrated that PEPD promotes the progression of glioblastoma through ZFP36-mediated alternative splicing of Insig1.
- Independently reviewed literature, formulated hypotheses, designed and carried out research projects.
 - Validated the PEPD–ZFP36–Insig1 axis using RNA-seq, qPCR, Western blot, Co-IP, RIP.

PUBLICATIONS

- [1] **Lei Shen***, Zhourui Zhang, Pengfei Wu, et al. “Mechanistic insight into glioma through spatially multidimensional proteomics”. In: *Science Advances* 10.7 (2024), eadk1721. DOI: 10.1126/sciadv.adk1721.
- [2] Jingyi Yang, **Lei Shen***, Yuankun Cai, et al. “The Role of Coagulation-Related Genes in Glioblastoma: A Comprehensive Analysis of the Tumor Microenvironment, Prognosis, and Treatment”. In: *Biochemical Genetics* (Mar. 2025). DOI: 10.1007/s10528-025-11086-3.
- [3] Jingyi Yang, **Lei Shen***, Jiabin Zhou, et al. “A Novel Mitochondrial-Related Gene Signature for the Prediction of Prognosis and Therapeutic Efficacy in Lower-Grade Glioma”. In: *Biochemical Genetics* (Oct. 2024). DOI: 10.1007/s10528-024-10928-w.

- [4] **Lei Shen*** and Nan Xiong. “Letter: Evaluation of 2 Surgical Techniques—Transposition Versus Interposition Microvascular Decompression for Hemifacial Spasm: A Systematic Review of 19 437 Patients”. In: *Neurosurgery* (May 2025). Epub ahead of print. DOI: 10.1227/neu.0000000000003496.
- [5] **Lei Shen***, Mengyang Wang, Jingwei Zhao, et al. “Study on the relationship between obesity and complications of Pediatric Epilepsy surgery”. In: *BMC Pediatrics* 23.1 (Mar. 2023), p. 142. DOI: 10.1186/s12887-023-03948-9.
- [6] **Lei Shen***, Jingyi Yang, Runqi Cheng, et al. “Bridge-layered decompression technique for vertebral artery-involved hemifacial spasm: technical note”. In: *BMC Surgery* 24.1 (May 2024), p. 154. DOI: 10.1186/s12893-024-02415-1.
- [7] Keyu Chen, **Lei Shen***, Jingyi Yang, et al. “A nomogram based on clinical multivariate factors predicts delayed cure after microvascular decompression for hemifacial spasm”. In: *Neurosurgical Review* 47.1 (Jan. 2024), p. 44. DOI: 10.1007/s10143-024-02284-5.
- [8] **Lei Shen*** et al. “An adolescent patient with anti-N-methyl-D-aspartate receptor encephalitis with motor aphasia as the first symptom and complicated by peripheral nerve damage: A case report and literature review”. In: *Medicine (Baltimore)* 104.19 (May 2025), e42436. DOI: 10.1097/MD.00000000000042436.

RELEVANT TECHNICAL SKILLS

Wet-lab skills:

- Cell culture (staining, scratch assay, colony formation, CCK8, Transwell).
- Molecular techniques (qPCR, Western blot, Immunohistochemistry, ELISA, Co-IP, ChIP, RIP).
- Basic animal handling, sampling, and behavioral testing.

Dry-lab skills: R, Python, bioinformatic analysis for bulk and single-cell RNA-seq, proteomics, metabolomics.

Clinical skills: Lumbar puncture, continuous lumbar subarachnoid drainage, and basic surgical skills.

Languages: Chinese (native), English (proficient)